

altairsim — examples

2026-08-10 · b27aff8

Machines that boot. Each directory here is self-contained: a `.toml` that describes the machine, the media it needs lying beside it, and a note saying what you will see. Copy any one of them anywhere and it still runs — a path inside a machine file resolves against **that file**, not against the directory you launched from.

```
altairsim examples/cpm/cpm22-buffered.toml    # CP/M 2.2b on an 8" floppy
altairsim examples/cpm/cpm22-terminal.toml    # ...the same, console in a built-in VT100 window
altairsim examples/basic/basic4k.toml         # Altair 4K BASIC, off a 1975 cassette
altairsim examples/diskbasic/diskbasic.toml   # Altair Disk Extended BASIC 4.1 on an 8" floppy
altairsim examples/uio/uio.toml               # Altair 8K BASIC over ONE 88-UIO (serial +
cassette)
altairsim examples/tarbell/tarbell.toml       # CP/M that boots itself from a Tarbell PROM
altairsim examples/hdsk/hdsk.toml             # CP/M off an 88-HDSK hard disk
altairsim examples/turnkey/floppy.toml        # an 8800bt Turnkey Module, no front panel
altairsim examples/cdos/cdos.toml             # Cromemco CDOS 2.58 off a 16FDC
altairsim examples/sdsys/sbc200.toml          # the SD Systems SBC-200 Z80 single-board computer
altairsim examples/sol/trek80.toml            # a Sol-20 at S0LOS, with TREK80 in the deck
altairsim examples/debugger/debugger.toml     # a bench for learning the symbolic debugger
altairsim examples/dazzler/kscope.toml        # a Cromemco Dazzler, drawing a kaleidoscope
altairsim examples/printing/printer.toml      # an 88-C700 printer, wired to a real host printer
altairsim examples/frontpanel/fp.toml         # the base machine, wired to the graphical front
panel
```

	What it is
<code>cpm/</code>	Mike Douglas's track-buffered CP/M 2.2b v2.3 , 56K, booted by the DBL PROM from an 8" floppy. <code>A></code> in one command. <code>cpm22-terminal.toml</code> boots the same disk with its console in a built-in VT100 window the simulator draws itself — no telnet client, no external emulator.
<code>basic/</code>	Altair 4K BASIC 3.1 (<code>basic4k.toml</code>) read off a period <code>.tap</code> by the bootstrap MITS shipped, unmodified: <code>MEMORY SIZE?</code> . Beside it, 8080 BASIC VER 1.0 (<code>basic1.toml</code>) — the first Altair BASIC, 1975 — booted the raw two-step way its primitive bootstrap forced: <code>RUN 1800</code> , <code>^E</code> , <code>RUN 0</code> → <code>MEMSIZ?</code> .
<code>diskbasic/</code>	Altair Disk Extended BASIC 4.1 (MITS, 1977) on an 8" floppy behind an 88-DCDD, booted by the DBL PROM. Unlike the cassette BASIC, this one has a filesystem — <code>SAVE</code> by name, a directory, <code>DSKINI</code> . <code>MEMORY SIZE?</code>
<code>uio/</code>	Altair 8K BASIC 3.2 over a single 88-UIO — the board that is a 6850 serial port (at 0x10) and an 88-ACR cassette (at 0x06) in one. The period bootstrap runs unmodified. <code>MEMORY SIZE?</code>
<code>tarbell/</code>	The Tarbell single- and double-density floppy interfaces, which carry their own 32-byte boot PROM: power on with a disk in the drive and 48K CP/M 2.2 comes up — no monitor, nothing to type.
<code>hdisk/</code>	CP/M 2.2 booting off an Altair 88-HDSK "Datakeeper" hard disk — an outboard controller moving whole sectors over an 88-4PIO — loaded by the HDBL PROM. <code>A0></code>
<code>turnkey/</code>	The MITS 8800bt Turnkey Module : an Altair with no front panel that boots itself the moment it powers on. Two files boot CP/M two ways — off an 88-DCDD floppy and off an 88-HDSK hard disk.
<code>cdos/</code>	Cromemco CDOS 2.58 , Cromemco's CP/M work-alike, booted from an 8" diskette through a Cromemco 16FDC — the board carrying the disk controller, console UART, and boot PROM at once.
<code>sdsys/</code>	The SD Systems SBC-200 , a whole Z80 computer on one S-100 board (8251 console, Z80-CTC, boot PROMs) running the MSMONR21 monitor; with a VersaFloppy it boots SDOS or CP/M.
<code>sol/</code>	A Processor Technology Sol-20 running SOLOS 1.3, with the 1977 game TREK80 on cassette. Type <code>XE TRK80</code> .
<code>debugger/</code>	A 46-byte program with its symbols and a guided walk through the monitor's debugger: <code>SYMBOLS LOAD</code> , symbolic <code>DISASM</code> , single-step, break on a label, run.
<code>dazzler/</code>	A Cromemco Dazzler , the S-100's first color graphics card, running Li-Chen Wang's Kaleidoscope . Comes up drawing a four-way-mirrored pattern in a window; <code>ATTN</code> (Ctrl-E) breaks back to the monitor.
<code>pmmi/</code>	A PMMI MM-103 modem and a small 8080 terminal program that bridges the console to the phone line. Four control keys work the modem — <code>^D</code> DTR, <code>^S</code> the 6860 Self Test loopback, <code>^I</code> modem status, <code>^C</code> quit. <code>^D</code> then <code>^S</code> makes the card echo your keystrokes with no phone line attached; set <code>dial / answer</code> to place a real call over TCP.

	What it is
<code>printing/</code>	An 88-C700 line printer , and a banner program that prints through it. The README sets up a real printer on your host — a network printer over <code>socket:</code> , or a CUPS queue over <code>printer:</code> — and a page comes out. Per-OS host setup (macOS, Linux; Windows pending).
<code>frontpanel/</code>	The base Altair with its front panel connected to altairsim-fp — a <i>separate</i> program (its own repo) that draws the panel's lamps and switches in a window and listens on TCP 8800 . Loads to the <code>altairsim></code> prompt; the window mirrors the running machine, and the on-screen sense switches feed a guest <code>IN 0FFH</code> .
<code>ai-mcp/</code>	A working directory for an AI assistant driving altairsim over MCP : a CP/M machine and a tiny <code>HELLO.ASM</code> with one deliberate bug the assistant assembles, runs, single-steps to find, and fixes — all through the simulator's MCP tools. See <code>DRIVING-WITH-AI.md</code> .

This tree is the product, which is the reason it exists as one directory rather than as media scattered through `disks/` and `tapes/`. It is what `tools/build-package.sh` assembles, it is what `docs/manual/quick-start.md` promises, and `acceptance-examples` boots every one of these out of a scratch directory with no repository in sight — because "does it work here" and "does it work where we hand it to people" turned out to be different questions.

Nothing is duplicated. `disks/` and `tapes/` keep only what does **not** ship: the period `.ASM` listings the boards were built from, download stubs for optional images, and vendor documentation.